

EnglishTalk - Free Version Application

Documentation

App Overview

EnglishTalk is a 100% free AI-powered English speaking practice app that helps users improve conversational English through real-time voice chats, pronunciation feedback, and interactive lessons.

Mission

Make English speaking practice accessible, fun, and effective for everyone - completely free.

Key Differentiator

Unlike other apps that require paid subscriptions, EnglishTalk uses free AI services and open-source tools to provide premium features at zero cost.

Core Features

1. AI Conversation Partner (FREE)
 - Chat with AI tutor on various topics
 - Real-time voice recognition and response
 - Adaptive difficulty based on user level
 - Multiple conversation scenarios (travel, business, daily life)
2. Pronunciation Coach (FREE)
 - Speech-to-text with accuracy scoring
 - Phoneme-level pronunciation feedback
 - Record and compare with native speakers
 - Common mistake detection
3. Text-to-Speech (FREE)
 - Natural voice playback for AI responses
 - Multiple voice options



- Speed and pitch control
- 4. Grammar Checker (FREE)
 - Real-time grammar correction
 - Spelling and punctuation check
 - Style and tone suggestions
- 5. Daily Lessons (FREE)
 - 5-10 minute quick lessons
 - Vocabulary building with context
 - Grammar in conversation
 - Idioms and phrases
- 6. Progress Tracking (FREE)
 - Speaking score dashboard
 - Streak and consistency tracker
 - Weak area identification
 - Weekly reports
- 7. Practice Modes (FREE)
 - Free Talk - Talk about anything
 - Role Play - Simulated real situations
 - Debate - Practice arguing opinions
 - Story Telling - Narrate stories aloud
 - Interview Prep - Job interview practice

Technical Architecture

Tech Stack (All Free)

- Frontend: React.js (Free, open source)
- Backend: Node.js + Express.js (Free, open source)
- Database: MongoDB (Free, MongoDB Atlas free tier 512MB)
- AI Chat: FreeTheAi or Completions.me (Free)
- Speech-to-Text: Browser-Whisper (Free, runs in browser)



- Text-to-Speech: Web Speech API (Free, built into browsers)
- Grammar Check: LanguageTool API (Free)
- Auth: JWT + Google OAuth (Free)
- Hosting: Vercel (frontend) + Railway (backend) (Free tier available)

System Flow

User Voice goes to Browser-Whisper (STT) then to FreeTheAi API (AI Processing) then to LanguageTool (Grammar Check) then to Web Speech API (TTS) then User Hears Response.

Cost Comparison

	PAID	FREE	SAVINGS
AI CHAT (OPENAI GPT-4)	\$100+/month	\$0 (FreeTheAi)	\$100+
SPEECH-TO-TEXT (WHISPER API)	\$50+/month	\$0 (Browser-Whisper)	\$50+
TEXT-TO-SPEECH (OPENAI TTS)	\$30+/month	\$0 (Web Speech API)	\$30+
GRAMMAR (GRAMMARLY)	\$12/month	\$0 (LanguageTool)	\$12
TOTAL	\$192+/month	\$0	\$192+

Project Structure

englishtalk folder contains:

- frontend folder - React.js Frontend
 - public folder
 - src folder with components, pages, services, hooks, context, utils, styles folders
 - App.jsx file
 - index.js file



- package.json file
- backend folder - Node.js + Express Backend
 - config folder
 - models folder for MongoDB Models
 - routes folder for API Routes
 - controllers folder for Request Handlers
 - services folder for Business Logic
 - middleware folder for Auth and Validation
 - utils folder
 - server.js file
 - package.json file
- docs folder - Documentation

Key Screens

1. Onboarding Screen

- Language level selection (Beginner, Intermediate, Advanced)
- Goal setting (Travel, Business, Casual)
- Daily reminder setup

2. Home Dashboard

- Daily streak counter
- Recommended lessons
- Quick practice button
- Progress summary

3. Conversation Screen

- Voice recording button
- Text transcript display
- AI response (text and audio)
- End conversation summary



4. Progress Screen

- Speaking accuracy chart
- Vocabulary learned
- Time practiced
- Achievements and badges

Database Schema (MongoDB)

Users Collection

- `_id`: ObjectId
- `name`: String
- `email`: String (unique)
- `password`: String (hashed)
- `level`: String (beginner, intermediate, advanced)
- `goal`: String (travel, business, casual)
- `streak`: Number (default 0)
- `lastPracticeDate`: Date
- `settings`: Object
- `createdAt`: Date
- `updatedAt`: Date

Lessons Collection

- `_id`: ObjectId
- `title`: String
- `description`: String
- `category`: String (vocabulary, grammar, conversation)
- `difficulty`: String (beginner, intermediate, advanced)
- `duration`: Number (minutes)
- `content`: Object
- `createdAt`: Date



Conversations Collection

- `_id`: ObjectId
- `userId`: ObjectId
- `topic`: String
- `mode`: String (freeTalk, rolePlay, debate, storyTelling, interview)
- `messages`: Array
- `overallScore`: Number
- `duration`: Number (seconds)
- `vocabularyUsed`: Array
- `grammarErrors`: Array
- `createdAt`: Date

Progress Collection

- `_id`: ObjectId
- `userId`: ObjectId
- `lessonId`: ObjectId
- `completed`: Boolean (default false)
- `score`: Number
- `timeSpent`: Number (seconds)
- `completedAt`: Date
- `attempts`: Number (default 1)

API Endpoints

Authentication

- POST `/api/auth/register` - Register new user
- POST `/api/auth/login` - Login user
- POST `/api/auth/google` - Google OAuth login
- POST `/api/auth/refresh` - Refresh JWT token



User

- GET /api/user/profile - Get user profile
- PUT /api/user/profile - Update user profile
- PUT /api/user/level - Update user level
- DELETE /api/user/account - Delete user account

Lessons

- GET /api/lessons - List all lessons
- GET /api/lessons/:id - Get lesson details
- POST /api/lessons/:id/complete - Mark lesson complete

Conversations

- POST /api/conversation/start - Start new conversation
- POST /api/conversation/message - Send message
- POST /api/conversation/end - End conversation, get summary
- GET /api/conversation/history - Get conversation history

AI Services

- POST /api/ai/chat - Get AI response
- POST /api/ai/grammar - Check grammar
- POST /api/ai/score - Score pronunciation

Progress

- GET /api/progress - Get user progress
- GET /api/progress/weekly - Get weekly report
- GET /api/progress/stats - Get detailed statistics



Free AI Services Used

1. AI Chat - FreeTheAi API

- Website: <https://freetheai.xyz>
- Models: 60+ free models including GPT-4 alternatives
- Limits: 250 messages/day (free tier)
- Setup: Sign up, get API key, use in app

2. AI Chat Alternative - Completions.me

- Website: <https://completions.me>
- Models: Claude Opus 4.6, GPT-5.2, Gemini 3.1 Pro
- Limits: Unlimited (according to website)
- Setup: Sign up, get API key, use in app

3. Speech-to-Text - Browser-Whisper

- Website: <https://www.npmjs.com/package/browser-whisper>
- How it works: Runs directly in user's browser
- Limits: Unlimited (runs on user's device)
- Setup: Install npm package, import in React

4. Text-to-Speech - Web Speech API

- Website: Built into all modern browsers
- How it works: Native browser API
- Limits: Unlimited (runs on user's device)
- Setup: No installation needed

5. Grammar Check - LanguageTool API

- Website: <https://languagetool.org>
- How it works: Free REST API
- Limits: 20 requests/minute, 20,000 characters/request
- Setup: No API key needed for basic use



Monetization Options (Future)

- Free Plan - \$0
 - 30 min/day
 - Basic lessons
 - All AI features
- Pro Plan - \$4.99/month
 - Unlimited practice
 - All lessons
 - Priority support
- Premium Plan - \$9.99/month
 - 1-on-1 AI tutor
 - Certificates
 - Offline mode

Environment Variables

Backend (.env file)

- PORT=5000
- MONGODB_URI=mongodb+srv://free-tier-cluster.mongodb.net/englishtalk
- JWT_SECRET=your_jwt_secret
- JWT_REFRESH_SECRET=your_refresh_secret
- FREETHEAI_API_KEY=your_freetheai_api_key
- GOOGLE_CLIENT_ID=your_google_client_id
- GOOGLE_CLIENT_SECRET=your_google_client_secret

Frontend (.env file)

- REACT_APP_API_URL=http://localhost:5000/api
- REACT_APP_FREETHEAI_KEY=your_freetheai_api_key
- REACT_APP_GOOGLE_CLIENT_ID=your_google_client_id



Free Hosting Options

- Vercel - React Frontend - Unlimited free tier
- Railway - Node.js Backend - \$5 credit/month
- MongoDB Atlas - Database - 512MB free
- Netlify - React Frontend alternative - 100GB bandwidth

Free Development Tools

- VS Code - Code editor - Free
- GitHub - Version control - Free
- Postman - API testing - Free

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